

SPORTZFY

Comprehensive Requirement Analysis & Market Validation Report

Course: Software Engineering (Sessional) — CSE-355
Department of Computer Science and Engineering (CSE)
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1. Executive Summary & Problem Overview

In Bangladesh, five-a-side and seven-a-side football turfs have become the default venue for casual and competitive matches among students, office groups, and local clubs, especially in cities like Chittagong and Dhaka. Despite high demand, the current booking workflow is manual, fragmented, and plagued by coordination friction.

This report evaluates and validates the requirements for Sportzfy — an on-demand real-time turf booking marketplace — by cross-referencing three sources of evidence:

- 22 social media community screenshots from groups such as Chittagong Turf Zone, Chittagong Turf Community, and Chittagong Turf Society
- A 25-respondent quantitative survey (Turf Booking App Feature Survey) from the CUET community and local players
- The Requirement Analysis Document and initial proposal

2. Ground-Truth Evidence: Qualitative Photo Analysis

Analysis of the 22 community screenshots reveals repetitive, severe bottlenecks that players and turf owners face daily:

Source Post	Real Problem Identified	Validated Feature
"Any turf free in CTG tonight at 8 PM?"	Players have no live visibility into open slots.	Live Real-Time Slot Availability Grid
"Late night (12–1 AM) 7v7 turf available?"	High demand for late-night slots and specific formats.	Time Slot Filters & Pitch Format Filters (6v6/7v7)
"Need a solid goalkeeper for tomorrow's match"	Squads are incomplete and must recruit strangers via Facebook.	Solo Player Matchmaking / Open Roster Hub
"Goalkeeper wanted, no age limit"	Individual players want casual football but lack a full team.	"Join a Match" Player Discovery Board
"Dominatrix FC — team match?"	Teams look for rival amateur teams to split turf cost.	Team vs Team Matchmaking Module
"Kazir Dewri Apollo turf 7v7? Slot price?"	Turf specs and pricing are hidden, forcing comment-thread queries.	Transparent Turf Specification & Pricing Card
Eco Sports: "Call or WhatsApp for booking"	Venue owners spend all day on manual calls/messages.	Owner Self-Service Management Dashboard
"Where to find good quality (master grade) turf?"	Players are skeptical of artificial grass quality.	Verified Player Reviews & Quality Badges

3. Quantitative Market Validation: Survey Analysis (25 Respondents)

Current Booking Behavior (Q3)

Response	Percent	Count
Always book via phone call	60.0%	15/25
Sometimes book via phone call	16.0%	4/25
Visit in person	16.0%	4/25
Use other apps	8.0%	2/25

Result: Over 76% of players are currently trapped in manual phone booking workflows.

Primary Selection Factors (Q2)

Response	Percent	Count
Availability / Timing	40.0%	10/25
Price	32.0%	8/25
Turf Quality	24.0%	6/25
Location	4.0%	1/25

Preferred Time of Day (Q4)

Response	Percent	Count
Night (8 PM onwards)	52.0%	13/25
Evening (5 PM–8 PM)	36.0%	9/25
Afternoon	12.0%	3/25

Result: 88% of all turf demand is concentrated in the evening and night.

Payment Security & Refund Hesitancy (Q5 & Q15)

Response	Percent	Count
Automated refund policy is “crucial”	72.0%	18/25
Willing to pay upfront only if cancellation policy is clear	52.0%	13/25

Result: Upfront-payment hesitation is tied directly to fear of losing money on weather or scheduling cancellations.

In-App Split Payment (Q6)

Response	Percent	Count
Extremely useful — we always split the bill	76.0%	19/25
Somewhat useful — nice to have	16.0%	4/25

Result: 92% positive response — essential feature for team sports.

Matchmaking & Opponent Finding (Q7)

Response	Percent	Count
Yes, frequently look for other teams	48.0%	12/25
Maybe occasionally	32.0%	8/25
Established friends only	20.0%	5/25

Result: 80% addressable user base for matchmaking.

1-Click Share Invites (Q11)

Response	Percent	Count
Very important — makes coordination much easier	76.0%	19/25
Somewhat important	20.0%	5/25

Result: 96% critical demand for WhatsApp/Messenger match invite sharing.

Equipment Rental Direct via App (Q8)

Response	Percent	Count
Yes, often forget or need extra gear	60.0%	15/25
Only for specific items like bibs	20.0%	5/25
Bring own	20.0%	5/25

Result: 80% willingness to rent gear (bibs, balls, goalkeeper gloves) during checkout.

AI Weather Forecasts & Rain Alerts (Q13)

Response	Percent	Count
Yes, weather ruins our plans frequently	60.0%	15/25
Nice to have	28.0%	7/25
Not needed	12.0%	3/25

Must-Have Facility Filters (Q14)

Response	Percent	Count
Clean washrooms and changing rooms	48.0%	12/25
Drinking water availability	32.0%	8/25
Spectator gallery	12.0%	3/25
Parking space	4.0%	1/25

4. Evaluation of the Requirement Analysis Document

The requirement analysis document accurately captured all survey statistics and established high-priority pillars. Key validations and refinements include:

- Validation of Pillars: Core Availability Engine, Split Payments, Matchmaking, Automated Refunds, and Amenities Filters are all verified with high statistical support.
- Refinement on Equipment Rental: the original proposal excluded rentals from course scope, but survey data shows 80% demand. Recommended compromise — a lightweight digital POS selector inside the booking workflow, leaving physical fulfillment to the venue.
- Refinement on Matchmaking: in addition to team-vs-team matches, the system should allow role-based recruitment (specifically for goalkeepers).

5. Feature Prioritization Matrix (MoSCoW Framework)

MUST HAVE	• Real-time slot locking engine (Hold & Release timer) • Search & Filter (Location, 6v6/7v7 format, Price, Time) • Digital Wallet Integration (bKash, Nagad) with Automated Refund Workflow • Turf Owner Management Dashboard (slot pricing & booking calendar)
SHOULD HAVE	• In-App Split Payment (divides total bill among players via payment links) • Matchmaking Hub (find opponents & recruit goalkeepers/solo players) • 1-Click WhatsApp/Messenger Share Invite (match time, location, split cost) • Equipment Rental Add-ons (bibs, match ball, gloves) • Facility Amenities Filter (washrooms, drinking water, seating)
COULD HAVE	• AI Weather Forecast & Rain Cancellation Alerts • Loyalty Points & Repeat Booking Discount System • AI Demand Prediction & Dynamic Pricing Suggestions for Owners
WON'T HAVE	• Physical gear inventory maintenance (managed on-site by turf owners) • Native mobile app (Android/iOS) — Next.js responsive web app for initial scope

6. Proposed Entity-Relationship & Backend Updates

To accommodate the validated features, the PostgreSQL backend schema should include:

Bookings Table Extensions

- split_payment_status: ENUM('none', 'pending', 'completed')
- equipment_rented: JSONB — e.g. {"bibs": 2, "ball": 1}
- cancellation_reason & refund_status: ENUM('none', 'requested', 'refunded')

Turfs Table Extensions

- amenities: TEXT[] — e.g. ['Washroom', 'Water', 'Parking', 'Seating']
- pitch_formats: TEXT[] — e.g. ['6v6', '7v7']
- quality_rating: DECIMAL(3,2)

SplitPayments Table (New)

- split_id, booking_id, player_phone, amount_due, payment_status

MatchRequests Table (New)

- match_id, host_user_id, turf_id, match_type (TeamVsTeam / PlayerRecruitment), required_role (Goalkeeper, Any), status (open, filled)

7. What Needs to Be Implemented & Technical Details

Based on the validated requirements and real-world user bottlenecks, here is the complete implementation breakdown across frontend, backend, database, and system workflows.

Module 1: Real-Time Turf Discovery & Dynamic Filtering Engine

- Live Availability Timetable Grid — interactive hour-by-hour matrix (4 PM to 2 AM), color-coded: Available / Locked-Pending Payment / Booked.
- Multi-parameter search & filter bar: Location/Area, Pitch Format (5v5/6v6/7v7), Time Window, Amenities Checklist, Price Range Slider.
- Turf Detail Profile Page: photo gallery, grass specifications, Google Maps embed, amenity badges, verified customer reviews.

Module 2: Concurrency Control & Instant Slot Locking (Hold & Release)

- 5-minute slot reservation lock: on “Book Now”, a temporary lock is acquired in Redis/PostgreSQL for 5 minutes (`lock_expires_at = NOW() + 5 minutes`).
- Real-time WebSocket sync (`/ws/turfs/{turf_id}/slots`): all viewers see a slot flip from Available to Locked instantly, no refresh.
- Automated Celery/background release worker: unpaid bookings auto-release the slot after 300 seconds.

Module 3: Digital Payment Gateway & Automated Refund Policy Engine

- Sandbox mobile wallet integration: bKash, Nagad, and Rocket payment APIs.
- Transparent automated refund system — 100% refund (>12h prior), 50% refund (6–12h prior), 0% refund (<6h prior).
- Severe weather / rainout guarantee: instant 100% refund credit when adverse weather is flagged or verified.
- Digital receipt & QR code confirmation for venue check-in.

Module 4: In-App Split Payment & 1-Click WhatsApp Invite Sharing

- Bill split calculator: captain enters team size; system computes per-player share and generates unique payment sub-links.
- 1-click WhatsApp/Messenger share button formats a match invitation template with date, time, location, and payment link.
- Live split tracking dashboard: shows the captain who has paid and who is pending in real time.

Module 5: Matchmaking Hub & Role-Based Player Recruitment

- Team vs Team (rival match) board: booked teams without an opponent post an open challenge to split the fee.
- Solo player & goalkeeper recruitment board: captains post vacant role-tagged spots; players set a profile (role, skill level, availability) and request to join.
- Instant push & WhatsApp notifications when a challenge is accepted or a squad member joins.

Module 6: Equipment Rental Checkout Selector (Digital POS)

- Gear checkbox add-ons at checkout: match ball (+150 BDT), bibs set (+100 BDT), goalkeeper gloves (+100 BDT).
- Turf owner equipment management: owners configure inventory counts and prices from their dashboard; rented gear attaches to the booking ticket.

Module 7: Turf Owner & Platform Admin Portals

Turf Owner Portal

- Self-service turf listing: pitch dimensions, photos, format, amenities.
- Custom slot pricing matrix: peak (8 PM–12 AM) vs off-peak pricing.
- Manual walk-in booking override to keep online inventory accurate.
- Revenue & payout ledger: earnings, withdrawal requests, daily occupancy.

Super Admin Portal

- Verification and approval of newly registered turfs.
- User management, dispute resolution, platform transaction analytics.

Module 8: AI Weather Alert & Intelligent Recommendation Layer

- AI/weather forecast API integration (OpenWeatherMap) using turf geolocation; displays real-time rain probability tags on slot cards.
- Proactive 3-hour rain warning when rain probability exceeds 70%, suggesting indoor alternatives or cancellation refund.
- Smart slot recommendation engine: heuristic/ML model on booking history, favorite pitches, and team sizes to recommend optimal times and venues.